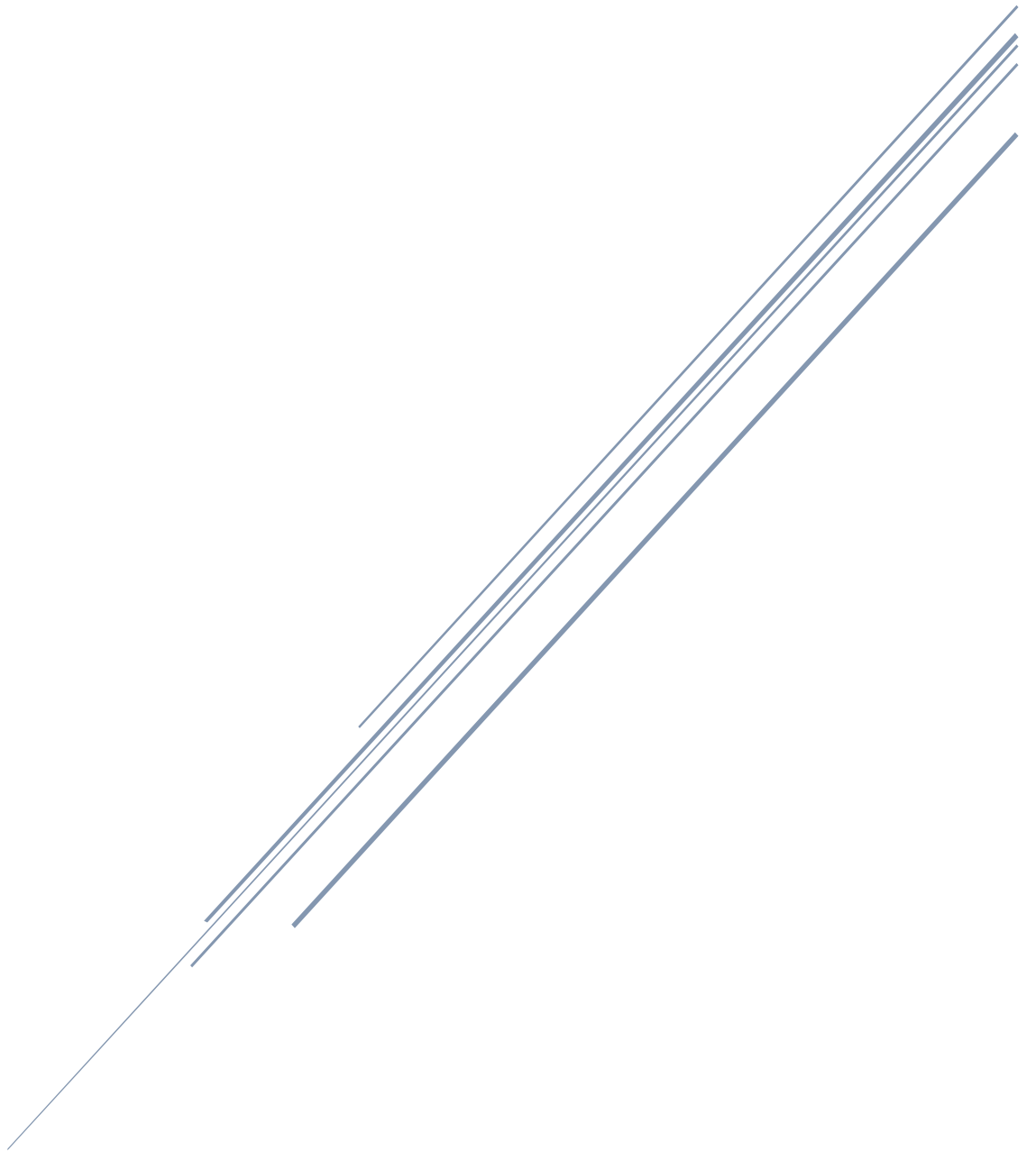


Time Series Modeling and Forecast Case Study Report



Student Name: Mouli Sirigiri
Student ID: 22022407

INTRODUCTION:

The time series data for Johnson & Johnson sales and the closing share price of Amazon are thoroughly analyzed in this study. We use RNN (Recurrent Neural Network) models (LSTM and GRU) and ARMA (AutoRegressive Moving Average) models to predict future trends. We offer insights into possible future trends and assess each model's performance through data pretreatment, time series analysis, model selection, and forecasting.

For comprehending and forecasting trends in sequential data, time series analysis is a vital tool. The two time series data sets that we are modeling and forecasting in this study are closing share price data from Amazon and sales data from Johnson & Johnson. We seek to produce precise predictions for the upcoming using both conventional statistical methods and cutting-edge deep learning techniques.

DATA DISCRIPTION

JJ SALES DATA

The Johnson & Johnson sales statistics is monthly sales information covering the period from January 1960 to December 1980. The relevant month and the sales amount, expressed in millions of dollars, are included in every data point. The dataset offers information about the company's sales success across a sizable time span for a number of product lines.

Synopsis:

Date: Indicates the sales data point's month and year.

Sales: Shows the overall quantity of sales for that specific month.

This information is useful for forecasting future sales performance, assessing seasonal patterns, and spotting long-term trends. Seasonality, trend, and possibly other underlying patterns in the data may need for extensive examination and modeling.

AMAZON SHARE PRICE DATA

Every day closing price of Amazon's stock is included in the closing share price data, which spans the years 2000 to 2023. The date and the closing price of Amazon's shares, expressed in US dollars, are included in every data point. It is possible to examine trends and generate forecasts using this dataset, which offers insights into the past performance of Amazon's stock.

Synopsis:

Date: This indicates when the trade day falls on.

Amazon's stock closing price for that trading day is indicated by the close price.

Making predictions for future changes in stock prices and assessing trends in the stock market are made easier with the help of this dataset, which indicates patterns like momentum and volatility. The market environment, the success of the company, and outside variables may all have an impact on the data's patterns.

METHODOLOGY:

1. Preparing data

A vital step in ensuring that the data is accurate, consistent, and prepared for analysis is data preparation. There are multiple steps involved here:

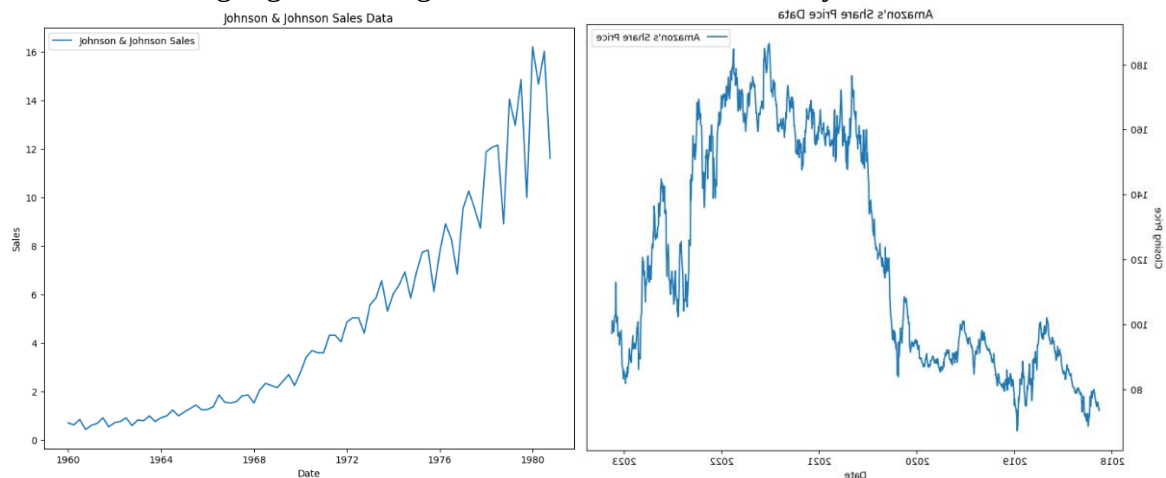
Handling Missing Values: Look for any missing values in the dataset and select a suitable response plan. Typical methods include removing rows or columns containing missing values or imputation, which replaces missing values with a reasonable estimate.

Date formatting: Make sure the date column is formatted properly and, if needed, change it to a datetime data type. This makes time-based graphing and analysis easier.

2. Examination of Time Series Data

In order to find patterns, trends, and seasonality in data, time series analysis looks at the data's temporal structure. Important actions in this stage consist of:

Visual Inspection: To visually examine the behavior of the time series data over time, plot the data. This can highlight recurring themes, trends, and seasonality.



Testing for Stationarity: Use statistical tests to ascertain whether the data shows seasonality, trends, or is stationary. One popular test is the Augmented Dickey-Fuller (ADF) test, which looks for unit roots in the data.

Transformation: To transform non-stationary data into stationary data, use techniques like differencing. In order to eliminate seasonality or patterns, differencing entails calculating the difference between two consecutive observations.

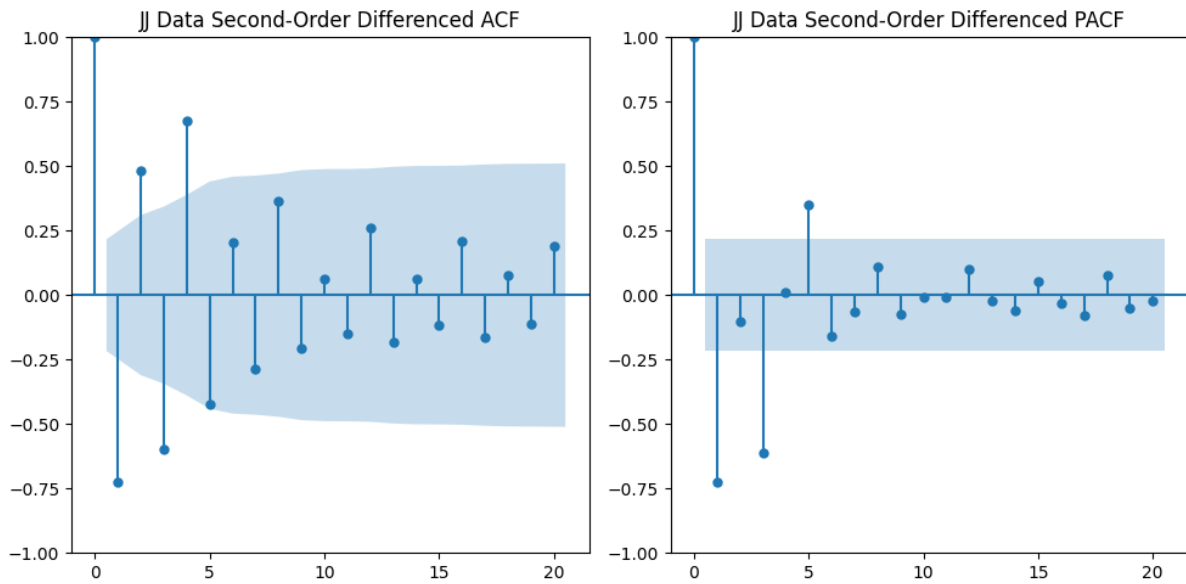
3. Modeling with ARMA

A class of statistical models called ARMA (AutoRegressive Moving Average) models is employed in forecasting and time series analysis. The moving average (MA) component of the ARMA model models the linear relationship between an observation and a stochastic term that represents random fluctuations, while the autoregressive (AR) component models the linear relationship between an observation and its lagged values.

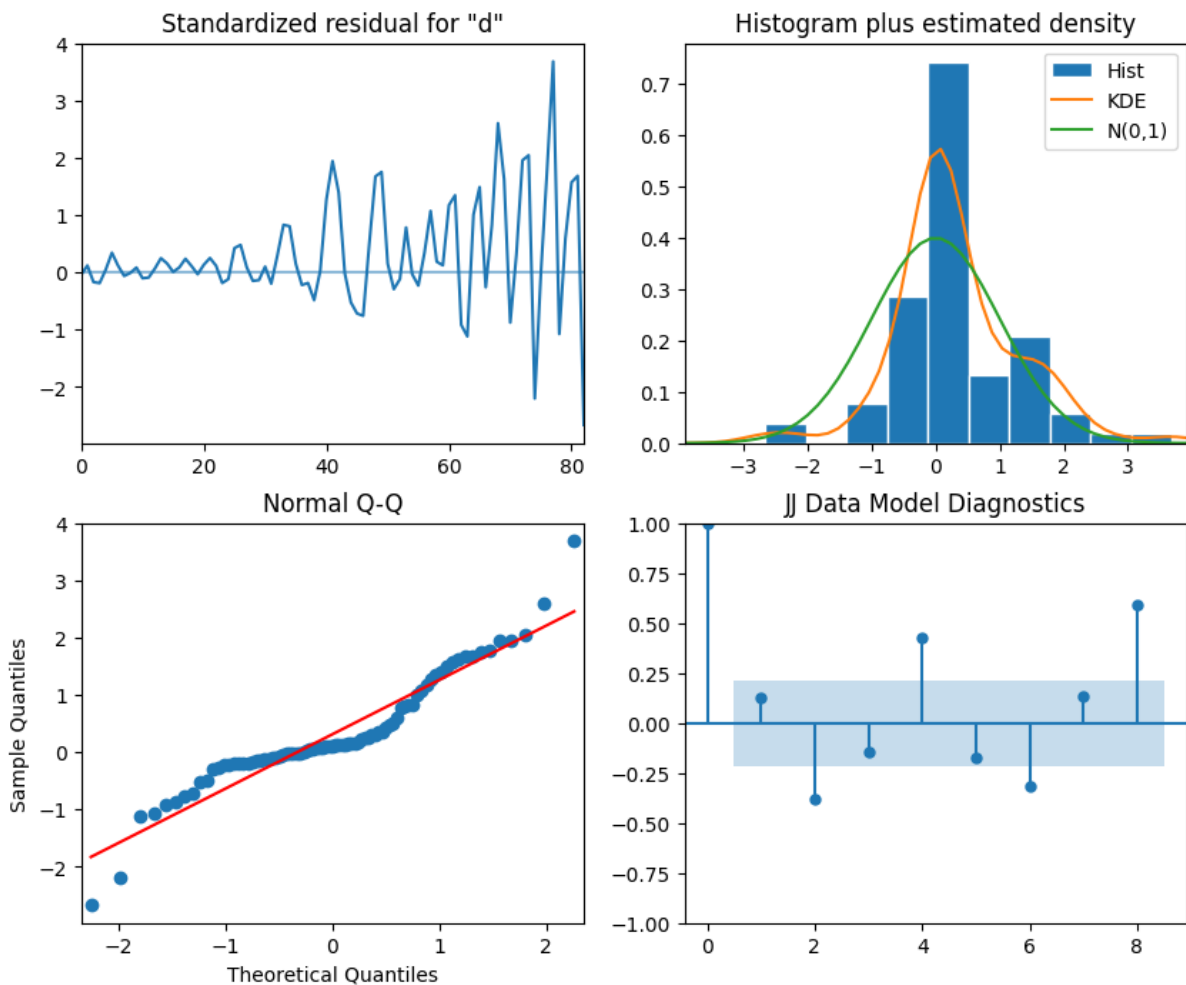
Important actions in ARMA modeling consist of:

Model Identification: To determine the order of the autoregressive (p) and moving average (q) terms in the model, use the autocorrelation function (ACF) and partial autocorrelation function (PACF) plots.

Parameter Estimation: Using maximum likelihood estimation or other optimization techniques, estimate the parameters of the ARMA model.



Model Evaluation: Utilizing information metrics like the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC), assess the fitted ARMA model. Using these standards, choose the model that fits the data the best.



4. RNN Models (GRU and LSTM)

Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) are two examples of recurrent neural network (RNN) models, which are deep learning architectures intended to

capture sequential dependencies in data. Time series forecasting problems are a good fit for these models.

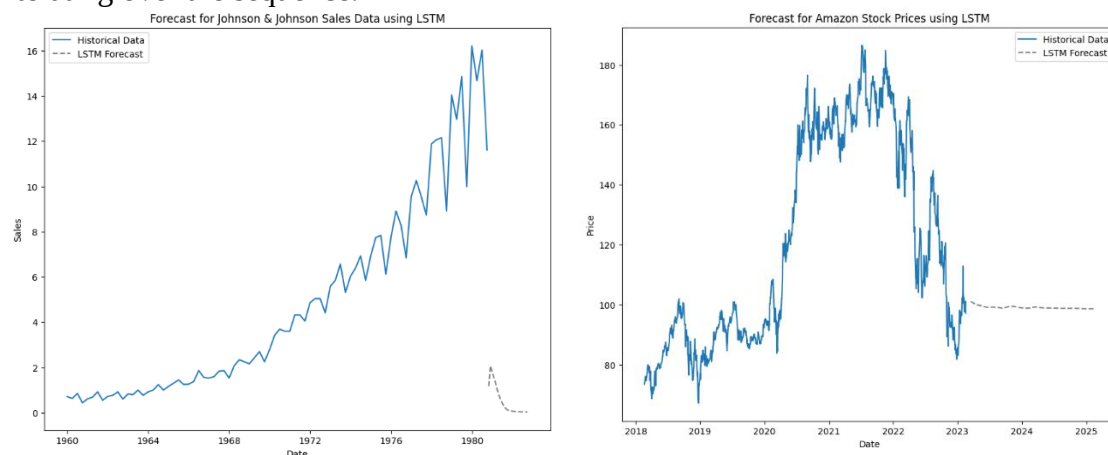
A few crucial stages in RNN modeling are:

Data Preparation: Sort the time series data into sequences that are appropriate for RNN model training by preprocessing it. The data may need to be scaled, input-output pairs made, and the data divided into training and validation sets.

Model Building: Describe the hidden units, activation functions, and number of layers in the LSTM or GRU model's architecture. Set up the model for data processing in a sequential manner.

Training: Using Adam or stochastic gradient descent (SGD), or another suitable optimization technique, train the RNN model using the training data. Observe the model's efficacy on the validation set in order to avoid overfitting.

Forecasting: Utilize the learned RNN model to produce forecasts for upcoming time steps after it has been trained. For the intended prediction horizon, construct multi-step forecasts by iterating over the sequence.



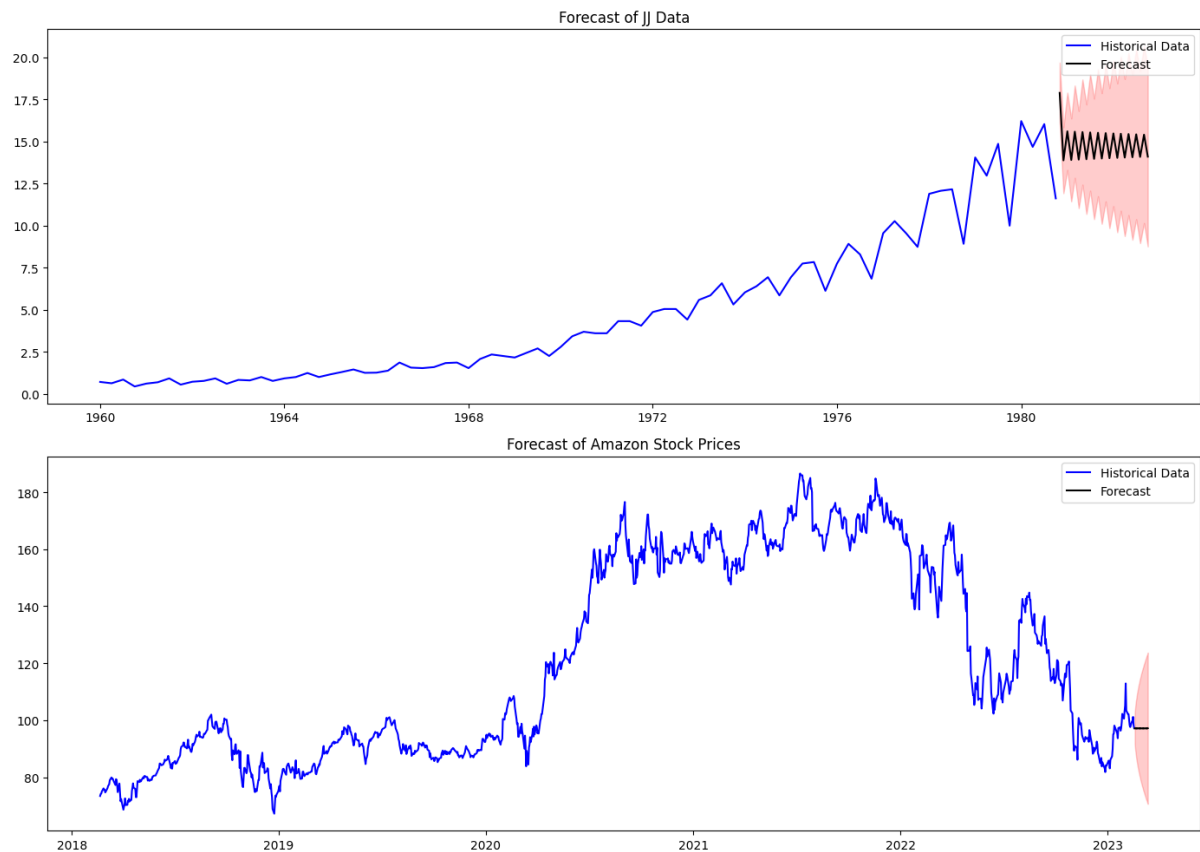
5. Predictive Analysis and Assessment

Once the ARMA and RNN models have been fitted, use each model to produce projections for the upcoming 24 months. Analyze the models' performance using suitable assessment metrics, such as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Squared Error (MSE). To choose the model that anticipates each dataset with the highest accuracy, compare the models' performances.

CONCLUSION:

Important insights into the past trends and possible future trajectories of these variables have been gained from the time series modeling and forecasting study performed on Johnson & Johnson sales data and Amazon's closing share price data. We have generated predictions for the next 24 months and developed a deeper understanding of the underlying patterns in the data by combining sophisticated deep learning techniques (RNN modeling) with conventional statistical methods (ARMA modeling).

The dynamics of Johnson & Johnson sales and Amazon's closing share prices have been better understood thanks to the time series modeling and forecasting study, in conclusion. Our projections, which can assist with investment strategies and strategic decision-making, are the result of our integration of cutting-edge deep learning algorithms with conventional statistical methodologies.



REFERENCES:

1. Jenkins, G. M., Box, G. E. P., and Reinsel, G. C. (2015). Time series analysis: control and forecasting. Wiley & Sons, Inc.
2. In 1997, Hochreiter, S., and Schmidhuber, J. extended short-term memory. 1735–1780 in Neural Computation, 9(8).
3. Bengio, Y., Bahdanau, D., Van Merriënboer, B., & Cho, K. (2014). Regarding neural machine translation properties: Encoder-decoder methodologies. Preprint arXiv:1409.1259.
4. Brooks, Charles (2008). Basic Econometrics for the Finance Sector. Cambridge University Press.
5. Athanasopoulos, G., and R. J. Hyndman (2018). Principles and applications of forecasting. OTexts.
6. J. D. Hamilton (1994). analysis of time series. Princeton University Press.
7. Bengio, Y., Goodfellow, I., and Courville, A. (2016). profound understanding. Press of MIT.
8. In 2019, Nielsen and Hansen published a book. Neural Networks and Deep Learning: A Textbook. Springer.
9. John Brownlee (2018). Deep Learning for Forecasting Time Series. Mastery of Machine Learning.